

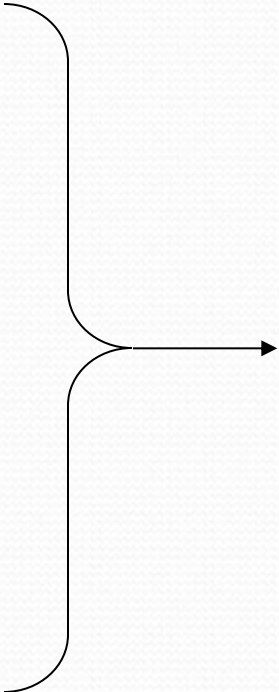
Building Java Programs

Advanced `if/else`

Factoring if/else code

- **factoring:** Extracting common/redundant code.
 - Can reduce or eliminate redundancy from if/else code.
- Example:

```
if (a == 1) {  
    System.out.println(a);  
    x = 3;  
    b = b + x;  
} else if (a == 2) {  
    System.out.println(a);  
    x = 6;  
    y = y + 10;  
    b = b + x;  
} else { // a == 3  
    System.out.println(a);  
    x = 9;  
    b = b + x;  
}
```



```
System.out.println(a);  
x = 3 * a;  
if (a == 2) {  
    y = y + 10;  
}  
b = b + x;
```


The "dangling if" problem

- What can be improved about the following code?

```
if (x < 0) {  
    System.out.println("x is negative");  
} else if (x >= 0) {  
    System.out.println("x is non-negative");  
}
```

- The second if test is unnecessary and can be removed:

```
if (x < 0) {  
    System.out.println("x is negative");  
} else {  
    System.out.println("x is non-negative");  
}
```

if/else with return

// Returns the larger of the two given integers.

```
public static int max(int a, int b) {  
    if (a > b) {  
        return a;  
    } else {  
        return b;  
    }  
}
```

- Methods can return different values using `if/else`
 - Returning a value causes a method to immediately exit.
 - All paths through the code must reach a `return` statement.

All paths must return

```
public static int max(int a, int b) {  
    if (a > b) {  
        return a;  
    }  
    // Error: not all paths return a value  
}
```

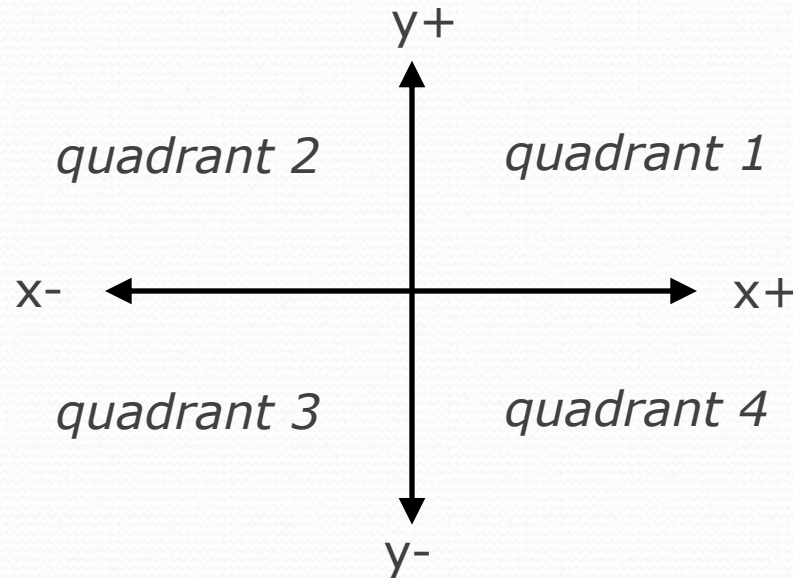
- The following also does not compile. Why not?

```
public static int max(int a, int b) {  
    if (a > b) {  
        return a;  
    } else if (b >= a) {  
        return b;  
    }  
}
```

- The compiler thinks `if/else/if` code can skip all paths, even though mathematically it must choose one or the other.
 - Solution here is to change `else if` to just `else`.

if/else, return question

- Write a method `quadrant` that accepts a pair of real numbers x and y and returns the quadrant for that point:



- Example: `quadrant(-4.2, 17.3)` returns 2
 - If the point falls directly on either axis, return 0.

if/else, return answer

```
public static int quadrant(double x, double y) {  
    if (x > 0 && y > 0) {  
        return 1;  
    } else if (x < 0 && y > 0) {  
        return 2;  
    } else if (x < 0 && y < 0) {  
        return 3;  
    } else if (x > 0 && y < 0) {  
        return 4;  
    } else {           // at least one coordinate equals 0  
        return 0;  
    }  
}
```